NEW DIRECTIONS IN **TRANSPLANTS**

These new medical breakthroughs are making organ transplants easier, with more successful outcomes

IMMUNOSUPPRESSIVE DRUGS



Immunosuppression medication like tacrolimus and cyclosporine suppress the immune system's response to a transplanted organ, helping its acceptance by the body

IMAGING FOR PRE- AND POST-SURGERY MONITORING



Before surgery, ultrasound, MRI and various CT scans help surgeons monitor patients. Post-surgery techniques like Ultrasound-Doppler show the state of the transplanted organ

MACHINE PERFUSION SYSTEMS



These devices circulate fluids through organs to mimic the body's blood flow, support organ viability during preservation and assessment

NEW PRESERVATIVES



New preservation solutions and techniques are being developed to minimise damage to organs during storage/ transportation

ROBOTIC SURGERY



Enhanced precision and minimally invasive techniques for transplant procedures help in faster recovery and better outcomes

AI DRIVEN ORGAN ALLOCATIONS



AI algorithms can analyse complex patient data to optimise donor–recipient matching, identifying the best possible organ for each individual

While most deceased donor organ transplants in India are carried out through Donation after Brain Death (DBD), organ Donation after Circulatory Death (DCD)—after circulatory and cardio-respiratory failure—is being increasingly used in the rest of the world. "In DCD, we declare death once the heart has stopped for five minutes and then have protocols in place for rapid retrieval to prevent damage to organs. The retrieved organs may be placed in machine perfusion systems (which ensure the artificial passage of blood, or 'perfusion', through the organs) where their function is assessed before implantation," explains Dr (Col.) Seth.

THE TECH EDGE

To meet the demand for organs, scientists are working on increasing availability from genetically modified animals, particularly pigs, trying to create bioengineered organs using 3D printing technology, and banking on stem cell research with an aim to regenerate damaged tissues/ organs.

"Innovations such as laparoscopic donor nephrectomy have made kidney donation less invasive, reducing post-operative recovery time. Immunosuppressive drugs have evolved, lowering the risk of organ rejection," says Dr Arun Kumar Balakrishnan, MD, Asian Institute of Nephrology and Urology (AINU) in Chennai. Advancements in robotic-assisted transplants are ensuring greater precision and minimal error. Using these techniques, Medanta can perform multi-organ transplants in a single surgery and can undertake liver transplants for advanced liver cancer, a procedure that a decade ago was found in medical fiction.

In another leap forward, there is the possibility for re-transplants—the process of replacing a previously transplanted organ with a new one—in case a transplanted organ creates problems. In 2023, a heart re-transplant was done in Karnataka. Hospitals like Apollo are also doing their bit to raise awareness, and have been setting up special teams to counsel patient families. Likewise, Manipal has the only cornea donation programme in India. The government, too, has taken the initiative to increase the availability of donor organs. As of 2025, this includes more public awareness campaigns, setting up a national eye bank registry for cornea transplants and an organ transport policy. Organs are being transported faster than ever before, with state-run and hospital-run green corridors.

More than anything else, awareness remains key in improving organ supply. Though Indians have their hearts in the right place—a 2018 survey by market research firm IPSOS found 74 per cent wanting to donate organs after death—lack of awareness of the process and age-old custom hold most back. "While medical advancements continue to enhance transplant success rates, increasing donor availability remains a collective responsibility," says Dr Navinath M., nephrologist at AINU. Only then can thousands get a second chance at life. ■